

ILMAAN ZIA

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Experienced Software Developer with 3 years of experience in backend development, performance optimization, and building scalable solutions. Skilled at delivering high-quality software solutions. Enthusiastic team player with a passion for continuous learning, eager to join a dynamic team in a challenging role.

Technical Skills

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| • Programming Languages, Frameworks - | Python, C++, Java, Django, FastAPI, DRF, Flask |
| • Front-End - | HTML, CSS, Javascript, Ajax, JQuery, React, |
| • Databases - | Postgres, SQL, SQLite, Elastic Search, Kibana, NoSQL, MongoDB, DBMS |
| • DevOPS & Version Control and Editor - | Heroku, AWS, Docker, PythonAnywhere, CPanel, GCP, Git, Vim, VSCode |
| • Data Visualization - | Tableau, Excel, Power BI, SQL, Google Analytics |

Work Experience

Research Assitant, College of Engineering,CSULB, California

From June 2023

- CSULB Student Summer Research Award (2024). Authored paper at QRS 2023 conference.
- Developing AI-driven security solutions for robotic systems, focusing on vulnerability detection and mitigation for local systems
- Fine-tuned Large Language Models on local GPU to enhance the detection of code vulnerabilities, improving overall system security.
- Utilized Langchain and Haystack for developing agentic workflows, and employed HyDE, MMR, and LLM reranking for effective semantic search. Fine tuned and compared Llama3.1 and Mistral Large Language Models on custom crafted datasets
- Performed Formal Modelling and VerificationTesting of robotic arm system. Solving and reducing unexpected test results causing system failures.
- Incorporating generative AI techniques to identify and address potential security risks in robotic systems.
- Contributing to the advancement of AI applications in robotics and cybersecurity through innovative research and development.

Software Developer, Trade India Infocom LTD , Noida

From Dec 2021 To Jan 2023

- 2022 TradeIndia Hackathon Runner-up. Investigated customer-reported issues, Improved Customer end design, resulting improving customer end efficiency to manage and update product orders.
- Integrated Python and Java Backend with React and NextJs based frontend.Developed and maintained TradeIndia's Backend Development and CRM Portal, managing and developing WordPress plugins for the blog page.
- Managed PostgreSQL databases, optimized complex queries, and improved data access efficiency by 30% for databases with 5+ million registered users.
- Collaborated with internal and external teams on Django to FAST-API integration, enhancing efficiency and effectiveness.
- Assisted in Migration of existing backend modules from Java to Python
- Designed pipelined, and deployed backend APIs from the ground up using FAST-API on GCP and AWS platforms.
- Assisted in migrating projects to microservices architecture, improving system and reducing latency by 20%.
- Established CI/CD pipeline automating testing, security, and hot reload deployments achieving approx 80% code coverage with 1500+ automated and manual unittest cases. And containerized modules using Docker and Kubernetes

Software Developer, SBNRI ,Gurugram

From Jan 2021 To Dec 2021

- Developed and maintained 3 key SBNRI projects: SBNRI web app, SBNRI blog project, and SBNRI Backend APIs.
- Created the SBNRI Full-Stack web application from scratch using Django Framework.
- Developed Frontend From Scratch using HTML, CSS and JavaScript Fetching frontend from Figma
- Automated form filling, built OCR Model to extract features from passport to autofill data to multiple forms.
- Architected and implemented RESTful APIs and WebSocket integrations for real-time data exchange,achieving better response times.
- Launched responsive SBNRI website with 5K monthly visitors, from idea to launch in 3 months using Django, jQuery, Bootstrap.
- Maintained and enhanced modules for the SBNRI blog project(Wordpress Based Front-end), Reducing loading time by 2X through front-end optimization of 15+ modules.
- Built REST APIs in Django Rest Framework for SBNRI mobile app, enabling mobile access for 3000+ users.
- Conducted regular code reviews and debugging sessions, enhancing code quality and reducing production issues.

Education

Master of Science in Computer Science,California State University, Long Beach, California

2023-2024

- Specializing in Advance Software Engineering and Advance Artificial Intelligence advised by Dr. Abdul Basit.
- Did Thesis Coursework on securing robotic systems using generative AI. Published research paper in QRS2023, SAM24, SERP2024.

Bachelors Of Technology Computer Science ,Jamia Hamdard, Delhi

2017-2021

- Contributed in Real world CNN application For Face Mask Detection For Covid-19 for image and video streams for CCTV cameras , [Link](#).
- Coursework: Advance Mathematics, Linux based systems, Java Spring, DBMS Systems,, Systems engineering, Data Structures, Algorithms.

Projects - [GitHub link- https://github.com/ilmaan](https://github.com/ilmaan) And Selected Research - [GoogleScholar](#)

- **Leveraging LLM To Detect And Correct Code Vulnerabilities**-Research Work and Developed Vulnerability Scanning App using Generative AI
 - Scraped vulnerability Dataset using beautifulsoup, preprocessed the dataset, fine tuned on mistral large language model from huggingface. Prompted the model to detect vulnerable patterns in code. Submitted Research accepted in **SERP2024 Conference**.
 - Developed web app to detect vulnerabilities in code, created chatbot application and file scanning application for report generation
- **Bug Tracking Application** - Bug Tracking Application Built with Django
 - Developed A Bug tracking application similar to JIRA for raising bug ticket and resolving project issues using CRUD operations.
- **Formal Modeling and Verification of Industrial Robotic Arm - A Case Study** - Published paper in IEEE Conference **QRS 2024**
 - Formally modeled and verified industrial robotic arm, ensuring safety and preventing deadlock using model-checking techniques.
- **Sign Language Recognition Model Using CNN, LSTM, and Google Mediapipe** - Research proposal to compare 2 deep learning models
 - Developed and evaluated advanced SLR models using CNN and Google Mediapipe with LSTM, achieving up to 95% accuracy.
- **People Solution Services** - Online Grievance Reporting Portal
 - Developed Django Web App for online grievance reporting, worked on Registration, Tracking, Submission and Feedback modules
- **Performance Analysis of Cache Size and Set-Associativity Using gem5** - Modeled and evaluated CPU cache configurations.
 - Highlighted differences in cache performance compared to SimpleScalar and implications for cache design.

Skills:

Unit Testing, Atlassian, Zendesk,Sharepoint, Full-Stack, Microservices, SDLC, ETL, Machine Learning, MS Excel, Jira, Bitbucket, Kubernetes, Shell Script, Agile Methodology, MVC, MVT, SCRUM, KANBAN, Waterfall , RESTful API, CDK, Lambda, Azure, GCP, automated testing, SaaS, UI Design, forecasting, GraphQL, design patterns, troubleshooting, webservises, software specification, requirement engineering, TypeScript, Redux, google cloud, data pipeline, computer vision, Kafka, CUDA